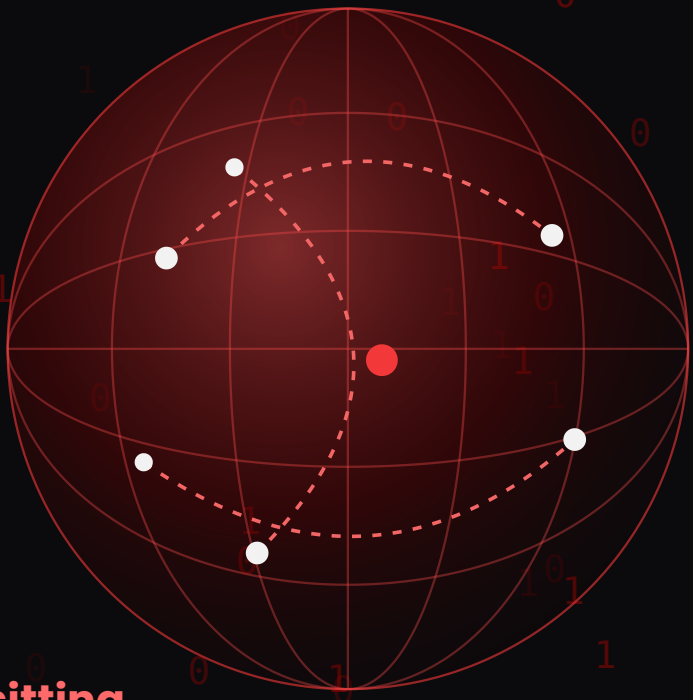
udyamdigital

YTL HACKATHON 2026 · ENTERPRISE TRACK A

Udyam Digital



Small businesses go digital in one sitting.

Approach, competitive analysis and impact of a phone-first journey for Accel Business: a voice advisor in 8 languages, one monthly price, KYC in minutes, a live tracker and a person on every exception.

7.4% → 11% lead-to-activation, baseline → pilot target	**17.6 → 8** days from lead to live	**192 / 192** automated tests pass	**0** ineligible bundles (15% today)

01 Executive summary
02 Problem & evidence
03 Approach
04 Build & architecture
05 Results
06 Competitive analysis
07 Impact
08 Privacy & next steps

Accel Telecom and Accel Business are the case's fictional operator. Project figures are computed from the supplied synthetic dataset and the project's own analysis and test runs; market facts are cited to public sources on the last page.

01 EXECUTIVE SUMMARY

A phone-first journey where rules decide, AI explains, and people handle exceptions.

Small businesses buying from Accel today wait for a call, make time for a visit, get a slow quote full of product codes, collect paper documents and then chase installation. Only 7.4% of leads become customers, after 17.6 days and 3 status calls each.

7.4%
of 5,000 leads become customers

17.6 days
from lead to live

Rs 15,053
to acquire one customer

3.0
status-chase calls each

What we built

- An AI advisor (text or voice, 8 languages) that asks at most 6 questions and returns a **rules-checked** bundle in plain words.
- An instant quote with add-on toggles and dependency locks: one monthly price.
- Smart KYC: DPDP consent first, live GSTIN/Udyam lookups, OCR on 4 documents and 13 cross-checks with specific fixes.
- A live tracker with expected dates, WhatsApp updates and automatic escalation to a named, language-matched agent.
- An admin dashboard: propensity-ranked leads, Call with outcome, and the 5 pilot metrics computed live.

Why it wins

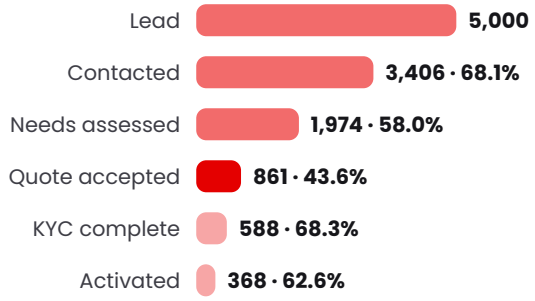
- **Aimed at the data.** 41.2% of drops are about reach and speed; the quote passes only 43.6%. The journey removes the visit and the wait.
- **Proven, not promised.** 192/192 tests, 15/15 judge-style scenarios, 12/12 KYC verdicts, a clean phone run from language to live.
- **Different from rivals.** Jio, Vi, Tata Tele and BSNL sell to MSMEs, but buying still runs through a callback, an assessment or a booking-then-wait.
- **Measurable.** At one quarter's 5,000 leads the pilot targets mean about 180 more customers and Rs 11.7 lakh less acquisition spend.

Recommendation: launch self-serve with professional offices (11.1% conversion, lowest CAC at Rs 10,695, 86% have a GSTIN) and run multi-site businesses as an assisted track from day one. Measure the five metrics weekly against the Apr–Jun 2026 baseline.

Two-thirds of losses happen before anyone sees a price.

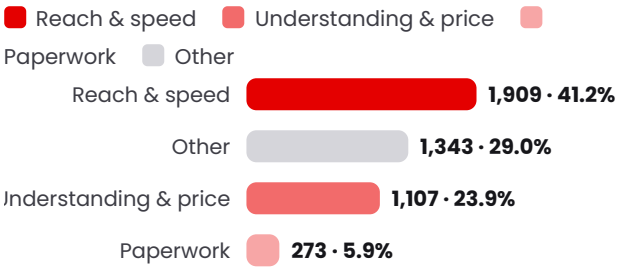
The case: design and build a digital-first journey that lets a small business owner discover the right bundle, buy it and get it activated with minimal human effort, on a phone, in their preferred language.

The funnel: 5,000 leads, Apr–Jun 2026



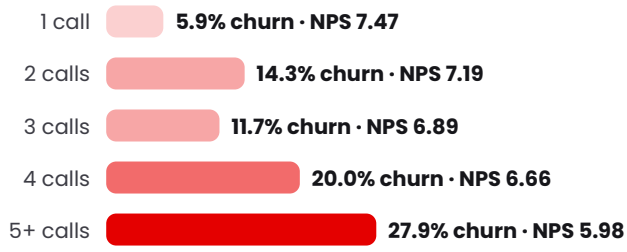
The largest loss is first contact (1,594); the lowest pass rate is the quote (43.6%, 32% for micro).

Why 4,632 leads dropped



Reach and speed cause 41.2% of drops; paperwork only 5.9%.

Chasing status predicts churn



Churn rises from 5.9% at one chase call to 27.9% at five or more: the tracker is a retention feature.

Other findings that shaped the design

- **Mis-selling:** 56 of 368 customers (15%) bought bundles the catalogue forbids.
- **GSTIN matters:** 9.6% conversion with a GSTIN vs 2.0% without; it gates 10 of 25 products, so the advisor asks for it at question 3.
- **Digital leads go unanswered:** 45% of website-form and 59% of WhatsApp leads are never contacted.
- **Lead scoring works:** the top 20% of scored leads hold 43.2% of activations.
- **KYC needs a safety net:** 6 of 12 sample applicants had planted problems; 2 need a person.

Win the segments that already convert, and keep micro shops self-serve.

Segment	Share	Converts	Median bundle	6-mo churn	CAC	Has GSTIN	Plan
Micro retail & services	54.2%	4.1%	Rs 2,470	22.7%	Rs 20,007	52.8%	Wave 2 · self-serve
Professional offices	24.9%	11.1%	Rs 6,575	13.8%	Rs 10,695	86.1%	Wave 1
Small multi-site businesses	15.1%	11.0%	Rs 28,080	10.8%	Rs 15,614	98.3%	Wave 1 · assisted
Small manufacturers	5.8%	12.8%	Rs 15,910	8.1%	Rs 15,324	99.3%	Assisted

Design principles: AI handles acquisition, people handle exceptions

Rules decide, AI explains
Eligibility, quantities and prices come only from code; LLM output naming an unlisted product or price is discarded.

People on exceptions
KYC judgement calls, late steps, a stuck chat or "talk to a person" go to a named agent.

Consent first
DPDP consent before any document; minimum data; IDs encrypted and masked.

Status, not chasing
Expected dates from catalogue install days, a live tracker and WhatsApp updates.

Field force included
Scored leads and an assisted mode, so sales and partners close more, not less.

The product decision: a hybrid of AI and people

Criterion	Weight	A · AI-first self-serve	B · Hybrid AI + human	C · Agent tool
Fit to the case (phone, language, field-force role)	25%	4	5	2
Addresses the observed drop causes	25%	3	5	3
Feasible within the hackathon	20%	4	3	4
Business value (CAC, conversion, churn)	15%	4	5	3
Risk (dead-ends, compliance, adoption)	15%	2	4	3
Weighted total (out of 5)		**3.45**	**4.45**	**2.95**

Scores are the team's judgement from the evidence. A dead-ends on exceptions (half the KYC sample needed a fix or review); C keeps the visit that causes 827 drops and the Rs 20–24k channel cost.

The journey: 11 screens on a phone

1. Pick a language
2. OTP login, WhatsApp opt-in
3. Six questions to the advisor
4. A rules-checked plan
5. One monthly price
6. DPDP consent
7. GSTIN registry lookup
8. Four documents, checked on upload
9. Verified, install booked
10. Tracker with the visit
11. Live, every update listed

Pink steps are where a named agent can take over with full context.

04 BUILD

A working prototype, built and tested phase by phase.

Phase	What was built	What it proved
1 · Core advisor	OTP login, 8 languages, rules engine with a reason for every refusal, 6-question advisor with live GSTIN checksum, guarded LLM explanations, recommendation card	10/10 profiles valid; 31 rule checks; no product codes on screen
2 · Instant quote	Pricing rule (micro at minimum, others at midpoint), toggles, steppers, dependency locks with reasons, "vs what you pay now", accept → order	Rs 1,597 and Rs 6,055 reference bundles match the analysis
3 · Smart KYC	Versioned consent in 8 languages, GSTIN checksum and registry, Udyam lookup, OCR with watermark removal, deskew and blur checks, 13 cross-checks, verdicts that move the order	12/12 verdicts match the planted problems; 0 raw IDs on screen or in logs
4 · Tracker & escalation	Order state machine, automatic install booking, SLA job, least-loaded language-matched assignment, callbacks inside working hours, agent console, order simulator	Late orders escalate unaided to the right agent
5 · Voice, WhatsApp, dashboard	Browser speech plus Sarvam/Bhashini adapters, short spoken replies; WhatsApp updates on 7 events + 24 h KYC reminder; propensity lead dashboard with Call and outcome; live metrics strip	Full Hindi voice chat; replies ≤ 180 characters; baselines reproduced live
6 · Hardening & deploy	15 judge-style scenarios, unit tests, Playwright happy path on a phone viewport, security pass (PII-safe logs, retention purge, rate limits, secrets), Docker + Render + Vercel configs	192/192 tests; 0 page errors on the phone run

Stack

Frontend

Next.js 15 and Tailwind, mobile-first, 8 languages (hi, en, mr, ta, bn, kn, gu, te), Web Speech for voice.

Backend

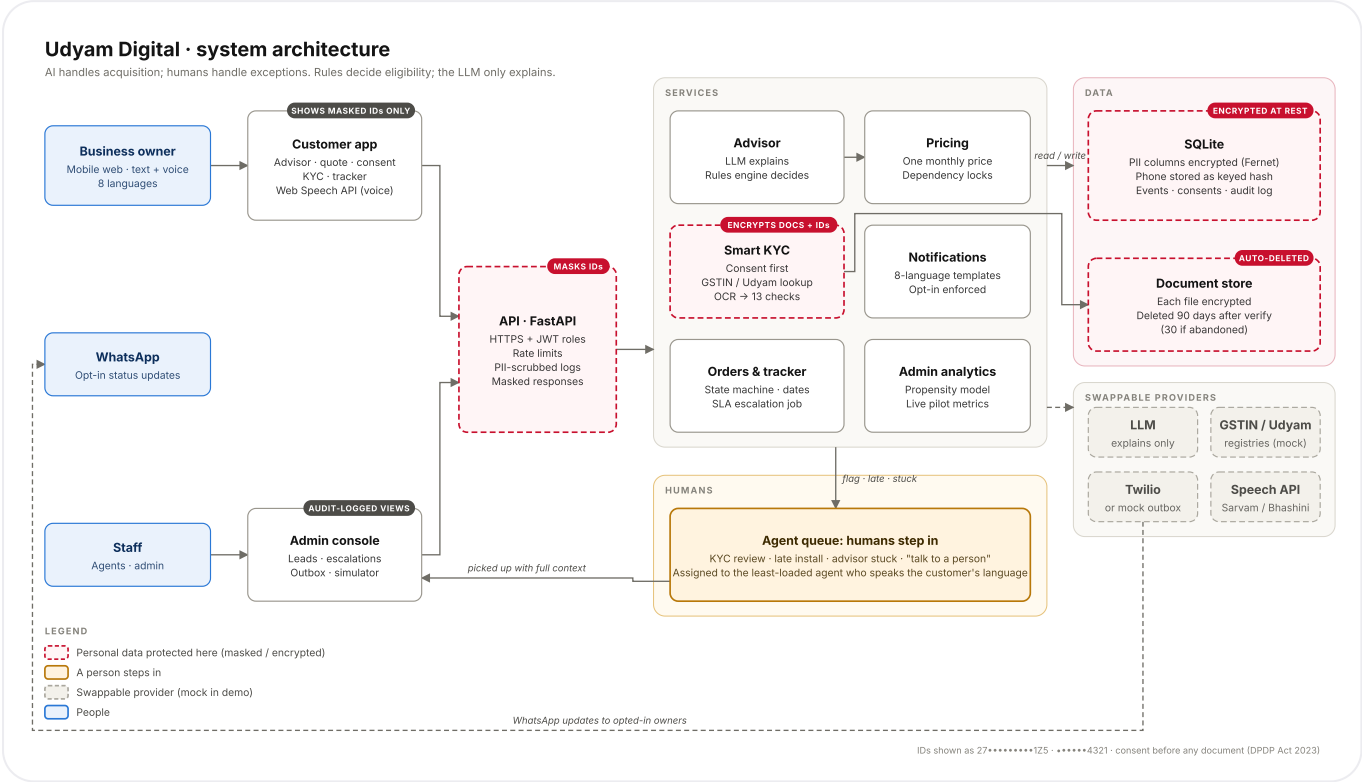
FastAPI, SQLAlchemy and SQLite, Tesseract OCR, scikit-learn propensity model, background SLA and notification jobs.

Providers

LLM, GSTIN/Udyam registries, Twilio WhatsApp and Sarvam/Bhashini speech, each behind one interface with a mock for the demo.

Demo: a 3-minute run from language to live on a phone, with a Hindi voice advisor, quote, KYC, tracker, an automatic escalation and the leads dashboard. The shot list and voiceover are in docs/demo_script.md.

One API, six services, one database, and a person in the loop.



Personal data protected. The API masks IDs in responses and logs; KYC encrypts documents and ID numbers; SQLite encrypts PII columns; documents auto-delete.

A person steps in. KYC review, late installs, a stuck advisor or "talk to a person" land in the agent queue with full context.

Swappable providers. LLM, registries, Twilio and Sarvam/Bhashini sit behind one interface each; mocks run the demo.

Main flows. Owner → customer app → API → advisor, pricing, KYC, orders; staff → admin console; updates go out on WhatsApp to opted-in owners.

05 RESULTS

Every claim is tested, including the judges' scenarios.

192 / 192
automated tests pass

15 / 15
judge-style advisor scenarios

12 / 12
KYC verdicts correct

0
page errors, language → live on a phone

15 advisor scenarios, typed the way a judge would

Scenario	Lang	Bundle	Monthly	Refused (with reason)
tier2 kirana no gstin	en	Dukaan Digital	Rs 2,196	WhatsApp Business API Starter
five site restaurant chain	en	Chain Connect	Rs 19,766	—
one person salon	en	Dukaan Digital	Rs 1,897	—
manufacturer wants iot	en	Factory Line	Rs 19,919	—
small factory iot ineligible	en	Factory Line	Rs 6,017	Machine Monitoring IoT
two shops want sdwan	en	Chain Connect	Rs 7,590	Managed SD-WAN
hindi medical store	hi	Dukaan Digital	Rs 2,196	—
cancelled gstin wants whatsapp	en	Dukaan Digital	Rs 1,796	WhatsApp Business API Starter
ca firm suspended wants tollfree	en	Office Pro	Rs 6,055	Toll-free Number
tiny office wants telephony	en	Office Pro	Rs 4,170	Cloud Telephony / SIP
coaching centre tier1	en	Office Pro	Rs 9,805	—
messy caps and typos	en	Office Pro	Rs 6,055	—
tamil chips textile unit	ta	Factory Line	Rs 16,675	—
marathi bakery add gst later	mr	Dukaan Digital	Rs 2,196	WhatsApp Business API Starter
gujarati three pharmacies	gu	Chain Connect	Rs 12,733	—

Smart KYC on the 12 sample applicants

K01 ✓ Blocked	K02 ✓ Verified	K03 ✓ Verified	K04 ✓ Re-upload needed	K05 ✓ Re-upload needed	K06 ✓ Human review
K07 ✓ Human review	K08 ✓ Verified	K09 ✓ Verified	K10 ✓ Verified	K11 ✓ Re-upload needed	K12 ✓ Verified

✓ = verdict matches the planted problem: 6 verified, 3 re-upload, 2 human review, 1 blocked.

Pilot baselines reproduce exactly

The admin dashboard recomputes the five metrics from the database: 7.4% conversion, 17.6 days, Rs 15,053 CAC, 15.2% churn and 3.0 chase calls, matching the Phase 1 analysis to the decimal.

The suite also checks that logs never contain raw GSTIN, Udyam, PAN or phone numbers, that uploads need consent, and that the API refuses to start without real secrets.

Phone run, language to live

A Playwright run on a Pixel 5 viewport completes all 11 screens with real uploads and no page errors, against a production-like backend.

Everyone sells to small businesses. Nobody sells the whole journey.

8.84 crore
enterprises on the Udyam portals, 30 Jun 2026 [1]

12%
of MSMEs fully digitally mature (DMI 58.0) [2]

< 3%
SME adoption of enterprise digital services vs 75–90% for large firms [3]

INR 954 bn
enterprise telecom opportunity through FY2030 [4]

The market is large, formalising fast and still mostly undigitised, which is why every Indian operator now runs an MSME programme. The question is how a business actually buys. From public information, the purchase still runs through a callback, an assessment or a booking followed by a wait; verification vendors sell components, not journeys.

Accel Telecom
The case's fictional operator (a top-2 telco).

Accel Business
Its B2B arm and the seller; today one product at a time through field sales.

Udyam Digital = our product
The digital journey Accel Business launches.

Capability comparison

Capability	Udyam Digital	Accel Business today	JioBusiness	Vi Business	Tata Tele	BSNL / ACT	KYC vendors
Self-serve purchase, start to finish	●	○	◐	○	○	◐	○
Guided discovery that ends in a bundle	●	◐	○	◐	◌	○	○
Eligibility rules enforced at sale	●	○	◌	◌	◌	◌	○
Instant, transparent quote with add-ons	●	○	◐	○	◌	◐	○
Digital KYC with GSTIN / Udyam checks	●	○	◌	◌	◌	◐	●
Live order tracker with expected dates	●	○	◐	◌	◌	◐	○
Vernacular voice in the buying journey	●	○	◌	◐	◐	○	○
Named person on exceptions, full context	●	◐	◐	◐	◐	○	○
Lead scoring for the sales team	●	○	◌	◌	◌	◌	○

● yes, publicly evidenced · ◐ partly · ○ not offered · ◌ not public. Assessed from public sources, not from inside each product, so competitor ratings may understate unpublicised features.

Competitor profiles

JioBusiness

high threat on price

Offer. Fibre plus productivity tools; seven plans from Rs 901 to Rs 10,001 at launch [5], now Business Broadband Max plans from Rs 801 with JioMeet and attendance tools [6].

How SMBs buy. Register interest and wait for an executive [7]; the page still leads with "Want a callback?", and the app manages accounts after purchase [8].

Our edge. Purchase and KYC finish in one session; the price is transparent and rules-checked.

Vi Business

advisory leader

Offer. Ready for Next: a free 10-minute digital maturity assessment with Dun & Bradstreet [9]; 1.9 lakh MSMEs assessed by Feb 2025 [10]; state MoUs with local-language training [11][12].

Gap. Diagnoses needs without turning them into a priced, eligible bundle.

Our edge. Six questions end in a plan with one monthly price the owner can buy immediately.

Tata Tele Business

closest strategic rival

Offer. 100% SMB-focused [13]; Smartflo cloud communications and a WhatsApp Business platform [14].

Direction. July 2026: a unified AI stack with Tata Communications, including voice agents and a digital B2B exchange [15].

Our edge. A working journey today, with KYC, tracking and an 8-language voice layer.

BSNL & ACT Fibernet

low threat on experience

Offer. BSNL takes bookings online or via WhatsApp "Hi" with document upload; an engineer calls within 3–7 days [16][17]. ACT sells city-specific SME plans [18].

Our edge. Bundle advice, an instant quote, expected dates and a live tracker.

IDfy · Signzy · HyperVerge

partners, not rivals

Offer. GSTN and Udyam verification, OCR and video KYC as APIs [19][20][21].

Our position. KYC checks sit behind one interface, so a vendor API replaces the mock registry without changing the journey.

The status quo

the real competitor

A visit, a hand-made quote, paper KYC and 17.6 days; 15% of sold bundles break catalogue rules; customers chase status 3 times.

Our edge. Every leak in the data has a feature aimed at it and a metric to prove it.

Positioning

	Advice before buying	How the purchase happens	After purchase
Operators (Jio, Vi, Tata Tele)	Plan pages; Vi's maturity assessment	Callback, sales executive or partner	Self-care apps and portals
Fixed-line ISPs (BSNL, ACT)	City plan lists	Online booking, then an engineer visit	Self-care portal
Udyam Digital	6-question advisor, eligible bundle with reasons	Instant quote, consent, KYC checked on upload, one session	Live tracker, WhatsApp updates, named agent when late

How rivals could respond, and our defence

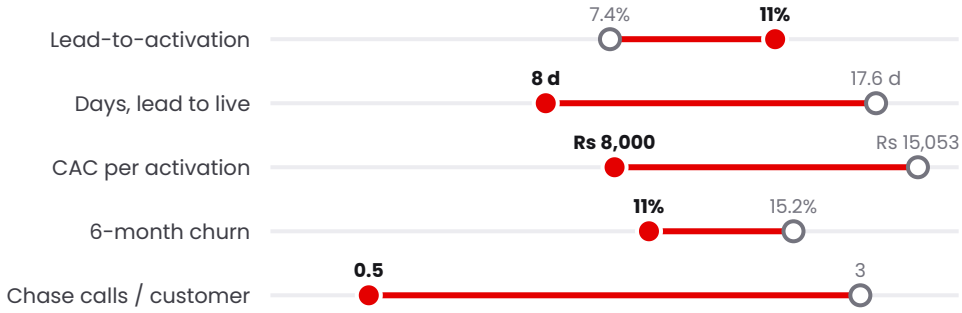
- **Price cuts (Jio).** Speed and clarity drive more drops (41.2%) than price (23.9%), and micro customers already get catalogue-minimum pricing.
- **AI agents (Tata).** Our moat is the deterministic rules engine and the KYC and escalation plumbing behind the conversation.
- **Advisory reach (Vi).** Route assessment-style traffic straight into the advisor so diagnosis ends in a purchase.

07 IMPACT

What a 6-month pilot could change, per quarter of leads.

Targets come from the data: professional offices already convert at 11.1%, referrals at 14.8%, and core installs take 5 days. They are pilot proposals, not results.

The five pilot metrics: baseline (Apr–Jun 2026) → 6-month target



What the targets mean in numbers

Per quarter	Today, 5,000 leads	Pilot targets, 5,000 leads	Pilot targets, 50,000 leads
Activations	370	**550** (+180)	**5,500** (+1,800)
Acquisition spend	Rs 55.7 lakh	**Rs 44.0 lakh**	**Rs 4.40 crore** (vs Rs 5.57 crore)
Status-chase calls	1,110	**275**	**2,750** (vs 11,100)
Extra customers still paying at 6 months	—	**+176**	**+1,757**
Added yearly revenue from them (× Rs 6,900)	—	**Rs 1.46 crore**	**Rs 14.55 crore**

Arithmetic on the stated targets and the dataset's baselines (7.4%, Rs 15,053, 3.0 calls, 15.2% churn, Rs 6,900 median bundle).

Proven in the build

now

- Zero ineligible bundles: checked by 30 rule tests and 25 advisor scenarios.
- One-session journey on a phone, language to live, with real uploads.
- Bad documents caught at upload with a specific fix.
- Late steps escalate unaided to the right agent.

To validate in the pilot

next

- Share of drops recovered: 32% (directly addressable) to 69% (including partly).
- A 50/50 self-serve / assisted mix, assisted deals 40% cheaper.
- KYC completion 68.3% → 85%; first response to escalations within 4 business hours.
- Whether vernacular voice closes the quote gap (40% regional vs 45%).

Beyond the numbers

For owners

No visit, no wait for a quote, one clear price in their language, and a date to check instead of calling.

For Accel Business

Higher conversion and lower CAC with zero ineligible sales and reproducible decisions.

For the field force

Time on multi-site deals, KYC exceptions and High-propensity leads, with credit for app-completed sales.

Collect the minimum, protect it, and delete it when the purpose ends.

The personal-data pipeline

1 Notice & consent
Plain notice in 8 languages (kyc-v1.0): what, why, how long, how to withdraw. No upload without it.
Login collects: phone, language, WhatsApp yes/no

2 Collect the minimum
Each step asks only what it needs. Voice audio becomes text and is discarded.
Advisor: type, size, city, needs; GSTIN optional

3 Verify
GSTIN / Udyam lookups, OCR and 13 cross-checks, used only for KYC.
KYC: GSTIN, Udyam, 4 documents

4 Encrypt & mask
Fernet on PII columns and every file; phone as a keyed hash; IDs masked in UI and logs.
Shown as 27•••••••••1Z5 · ••••••4321

5 Access by role
Customer sees own data; agent only assigned cases; admin the rest.
Every view is audit-logged

6 Retain
Kept only while the purpose runs. Cases under review are held until decided.
90 days after verify · 30 if abandoned

7 Delete
Hourly purge job. Withdrawal cancels the case and deletes files at once.
One tap in the app

Controls on every step

- HTTPS only
- Fernet-encrypted PII and files
- Phone as keyed hash
- Masked IDs in UI and logs
- Role-based access
- Audit log of every view
- Rate limits
- Secrets from env only

Maps to the DPDP Act 2023 · our reading, not legal advice

Choice	DPDP point
Notice and explicit consent before KYC	s.5 notice, s.6 consent
KYC used only for verification	Purpose limitation
Per-step minimisation	Only data necessary (s.6(1))
One-tap withdrawal with deletion	s.6(4), erasure (s.12)
Auto-deletion after the purpose ends	s.8(7)
Encryption, masking, access control, audit	Security safeguards (s.8(5))
Owner sees status and data; can ask for help	s.11, grievance (s.13)

Next steps

0–3 months · Pilot

- Professional offices in 2 tier-1 cities
- Real GSTN / Udyam lookups via a GSP or KYC vendor
- DigiLocker documents, Aadhaar e-sign
- Grievance officer and published notice

3–6 months · Scale

- WhatsApp Business API and real SMS OTP
- Postgres with keys in a cloud KMS
- Partner and field-force portal with shared credit
- Micro-retail wave, lean bundle

6+ months · Later

- DPDP consent manager integration
- Voice-first flow on WhatsApp
- Retrain propensity on live outcomes; bias checks
- Single bill and one support number

09 SOURCES

Sources

Project figures: the supplied synthetic dataset (smb_leads.csv, product_catalogue.csv, mock GSTIN and Udyam registries, 47 KYC images) and the project's analysis, test and end-to-end outputs (docs/kyc_results.md, docs/advisor_scenarios.md, the 192-test suite). Market and competitor facts:

1. KNN India: over 8.84 crore MSMEs registered on Udyam platforms (Jul 2026)
2. CRN India: Vi Business Ready for Next MSME study 2025 (DMI 58.0, 12% fully mature)
3. Multibagg: India's SMEs under-3% enterprise digital services adoption (Jio filing)
4. Frost & Sullivan: INR 954 billion enterprise telecom opportunity through FY2030
5. Disruptive.Asia: Jio launches seven plans, Rs 901 to Rs 10,001
6. TelecomTalk: Jio Business Broadband Max 801 plan (Mar 2026)
7. CIO Tech Outlook: register interest and wait for a JioBusiness executive
8. Jio: JioBusiness Solution page (callback form, self-care portal)
9. Vi Business: Ready for Next assessment (10 minutes, free)
10. Vi press release: 1.9 lakh MSMEs assessed by Feb 2025
11. TelecomTalk: Vi MoU with West Bengal, local-language advisory
12. TelecomTalk: Vi Business MoU with Andhra Pradesh (Jul 2026)
13. TelecomTalk interview: TTBS is 100% focused on SMBs
14. TTBS: end-to-end digital solutions, Smartflo, WhatsApp platform
15. NSE filing: Tata Communications and TTBS unified AI stack for SMBs (Jul 2026)
16. BSNL: book FTTH online or WhatsApp 'Hi' to 18004444
17. CitizenNest: BSNL online application, engineer contacts in 3–7 days
18. ACT Fibernet: business broadband plans
19. HyperVerge: GSTN verification
20. G2: IDfy KYC, KYB and 140+ verification APIs
21. Signzy KYB API: GSTIN, CIN, PAN verification

Interactive version: the Udyam Digital showcase website, with the demo video, live charts, the impact calculator and the architecture diagram.